

MSc Data Science

Project 7PAM2002

Department of Physics, Astronomy and Mathematics

FINAL PROJECT REPORT

Project Title:

UK house price prediction by using time-series analysis, Machine Learning and Deep Learning Models

[Project Git Hub Link](#)

Student Name: [Sridhar Guggilla](#)

Student ID: [23021710](#)

Supervisor: [Hyungrok Kim](#)

Date of Submission: [06/01/2026](#)

Word Count: [5907](#)

DECLARATION

This report has been submitted as part of the requirements for the award of the Master of Science in Data Science at the University of Hertfordshire. I confirm that I have reviewed and understood the University's regulations relating to academic integrity, including policies on plagiarism and academic misconduct, and I am aware of the procedures and potential consequences associated with breaches of these regulations.

I declare that this work is entirely my own and that all sources of information, whether published or unpublished, have been appropriately referenced and acknowledged within the report. No part of this submission has been copied or reproduced without proper attribution. This project did not involve the use of human participants or personal data.

I grant permission for this report to be made available for academic purposes through University platforms, provided that appropriate acknowledgement of the source is maintained.

Student Name: Sridhar Guggilla

Signature: [Sridhar. G](#)

Student ID: [23021710](#)

UNIVERSITY OF HERTFORDSHIRE
SCHOOL OF PHYSICS, ENGINEERING AND COMPUTER SCIENCE

Abstract

This study examines the forecasting performance of statistical, machine learning, and deep learning models when applied to long-term UK house price data. Monthly observations from the UK House Price Index covering the period 1995 to 2024 were used to evaluate whether increasingly complex models consistently provide superior predictive accuracy compared to simpler approaches.

A unified modelling framework was adopted to ensure fair comparison across methods, including identical preprocessing steps, chronological train–test splitting, and common evaluation metrics. Four forecasting models were implemented: ARIMA, SARIMA, Long Short-Term Memory (LSTM), and Linear Regression using lag-based features. Model performance was assessed using RMSE, MAE, MAPE, and R^2 to capture both prediction error and explanatory capability.

The findings show that while SARIMA and LSTM effectively represent seasonal structure and non-linear temporal behavior, a Linear Regression model with engineered lag features achieved the most accurate and stable forecasts across all evaluation measures. This outcome suggests that, for trend-dominated economic time series such as UK house prices, feature representation and methodological discipline may be more influential than algorithmic complexity. The study highlights the importance of structured evaluation when selecting forecasting models for economic time-series applications.

Table of Contents

1. Introduction.....	6
1.1 Research Problem.....	6
1.2 Research aim and objectives	7
1.3 Scope of Study	7
2. Literature Review	7
2.1 House Price Forecasting Approaches	8
2.2 Classical Time-Series Models	8
2.3 Machine Learning and Deep Learning Models.....	8
2.4 Regression-Based Forecasting and Feature Engineering.....	9
2.5 Research Gap	9
3. Dataset, Exploratory Analysis and Preprocessing	10
3.1 Data Source and Description	10
3.2 Exploratory Data Analysis.....	10
3.3 Distributional Properties.....	11
3.4 Data Preprocessing	11
4. Ethical Considerations	13
5. Methodology	13
5.1 Overview of the Modelling Framework	13
5.2 Time-Aware Train–Test Split Strategy	13
5.3 Model Development Model Development.....	13
5.4 ARIMA Model Implementation	14
5.5 SARIMA Model Implementation.....	14
5.6 LSTM Model Architecture and Training.....	14
5.7 Linear Regression with Lagged Features	15
6. Results	16
6.1 ARIMA Model Results.....	16
6.2 Augmented Dickey-Fuller test.....	16
6.3 SARIMA Model Results	17
6.4 LSTM Model Results.....	17

6.5	Linear regression	18
6.6	Hyperparameter Tuning of the LSTM Model	19
7.	Evaluation Metrics	20
7.1	Root Mean Squared Error (RMSE)	20
7.2	Mean Absolute Error (MAE).....	21
7.3	Mean Squared Error (MSE).....	21
7.4	R-squared (R^2)	21
8.	Analysis and Discussion	22
8.1	Comparative Model Performance Analysis.....	22
8.2	Discussion of Results in Relation to Literature	22
8.3	Interpretation of Forecasting Behaviour	23
9.	Conclusion and Future Work	23
9.1	Summary of Key Findings	23
9.2	Future Research Directions	23
10.	References	24
11.	Appendix	25

1. Introduction

House prices are a key indicator of economic conditions in the United Kingdom, influencing household wealth, mortgage lending, investment decisions, and government policy. Fluctuations in property values affect both financial stability and housing affordability, making accurate house price forecasting a subject of sustained interest for policymakers, financial institutions, and market participants. Reliable forecasts acknowledged not only support strategic planning but also assist in understanding broader market dynamics during periods of economic uncertainty.

Despite the availability of long historical records, forecasting house prices remains challenging. UK house price data exhibits strong long-term growth, recurring seasonal patterns, and high temporal dependence, while also being influenced by structural changes such as policy interventions and economic shocks. These characteristics result in non-stationary and autocorrelated time-series data, limiting the effectiveness of simplistic modelling approaches. Consequently, researchers have increasingly explored a wide range of forecasting techniques to better capture these dynamics.

Classical statistical time-series models, such as Autoregressive Integrated Moving Average (ARIMA) and Seasonal ARIMA (SARIMA), have traditionally been used in economic forecasting due to their transparent structure and solid theoretical foundation. More recently, machine learning and deep learning approaches, particularly LSTM networks, have gained prominence because of their ability to model non-linear relationships and long-term dependencies. However, the growing adoption of complex models has often been accompanied by limited critical evaluation against simpler alternatives under consistent experimental settings.

This project adopts a comparative approach to evaluate whether increasing model complexity necessarily leads to improved forecasting performance in the context of UK house prices. By analysing long-term monthly data from the UK House Price Index and applying multiple forecasting models within a unified framework, the study aims to provide empirical evidence on the trade-off between model sophistication, accuracy, and stability.

1.1 Research Problem

A significant challenge in UK house price forecasting lies in the lack of clear empirical consensus regarding the relative effectiveness of different modelling approaches. While recent studies frequently promote machine learning and deep learning models as superior forecasting solutions, many fail to benchmark these methods rigorously against classical statistical models or well-engineered regression approaches. Inconsistent preprocessing strategies, inappropriate validation techniques, and selective performance reporting further complicate the interpretation of reported results.

As a result, it remains unclear whether observed improvements in forecasting accuracy are genuinely attributable to model complexity or instead arise from methodological choices such as feature representation, data splitting strategies, or evaluation metrics. This ambiguity is particularly relevant for long-term, trend-dominated economic time series like UK house prices, where simpler models may already capture a substantial proportion of the underlying structure.

The core research problem addressed in this study is therefore the absence of a controlled and transparent comparison between classical time-series models, deep learning approaches, and regression-based forecasting methods when applied to long-term UK house price data. Addressing this gap is essential for identifying forecasting techniques that balance accuracy, interpretability, and practical applicability.

1.2 Research aim and objectives

The primary aim of this project is to systematically compare the forecasting performance of classical time-series models, deep learning models, and regression-based approaches in predicting UK house prices using long-term monthly data.

Objectives

The specific objectives of the study are to:

1. Analyze historical UK house price data to identify key temporal characteristics relevant to forecasting.
2. Apply appropriate preprocessing techniques to ensure suitability for time-series modelling.
3. Implement ARIMA, SARIMA, LSTM, and Linear Regression models using consistent training and testing strategies.
4. Evaluate and compare model performance using standard forecasting metrics.
5. Interpret differences in model behaviour and performance in relation to data characteristics and methodological choices.

1.3 Scope of Study

This study focuses on national-level average UK house prices using monthly data from 1978 to 2024 obtained from the UK House Price Index. The analysis is restricted to univariate time-series forecasting, where past house price values are used as the sole input for prediction. External macroeconomic variables are not incorporated. Model evaluation is conducted using a time-aware train–test split and standard accuracy metrics. The study is intended as a comparative assessment of forecasting approaches rather than a causal analysis or real-time deployment exercise.

2. Literature Review

House price forecasting is a well-established research area due to its importance in economic planning, investment decision-making, and financial risk assessment. Housing market data typically exhibits long-term trends, seasonality, and strong temporal dependence, making it

suitable for time-series forecasting while also presenting significant modelling challenges.

2.1 House Price Forecasting Approaches

House price forecasting has attracted sustained academic and professional interest due to its relevance to economic planning, financial stability, and investment decision-making. Property price data typically exhibits persistent upward trends, seasonal variation, and strong dependence on historical values, which makes forecasting feasible but methodologically challenging. As a result, a wide range of modelling approaches has been proposed, differing in their assumptions, complexity, and interpretability.

Broadly, existing research can be categorized into three groups: classical statistical time-series models, machine learning and deep learning approaches, and regression-based forecasting methods enhanced through feature engineering. Each group offers distinct advantages and limitations when applied to long-term economic time series such as house prices. Rather than focusing solely on predictive accuracy, recent studies increasingly emphasize the trade-offs between model performance, robustness, and practical applicability.

2.2 Classical Time-Series Models

Classical time-series models, particularly Autoregressive Integrated Moving Average (ARIMA) and its seasonal extension SARIMA, have been widely used as baseline forecasting tools in housing market analysis. These models rely on the assumption that future values can be inferred from linear relationships within historical observations, with differencing applied to address non-stationarity.

ARIMA models have demonstrated reasonable short-term forecasting capability in housing market studies, especially when the data exhibits stable patterns after transformation. However, their reliance on linear structure limits their ability to extrapolate long-term growth accurately, particularly in datasets dominated by strong trends. SARIMA extends this framework by explicitly modelling seasonal components, allowing periodic patterns such as annual housing cycles to be captured more effectively.

Despite their theoretical grounding and interpretability, several studies report that ARIMA and SARIMA models struggle when applied to long forecasting horizons or structurally evolving markets. Parameter sensitivity and the requirement for careful manual specification further reduce their flexibility. Consequently, while classical models remain valuable as benchmarks, their suitability for long-term house price forecasting is often constrained.

2.3 Machine Learning and Deep Learning Models

The limitations of classical statistical models have motivated increased use of machine learning and deep learning techniques in house price forecasting. Unlike traditional time-series methods, these approaches do not impose strict linearity assumptions and can learn complex, non-linear relationships directly from data.

Among deep learning architectures, LSTM networks have gained particular attention due to

their ability to model sequential dependencies through gated memory mechanisms. Empirical studies frequently report that LSTM models outperform ARIMA-based approaches, especially when the data contains non-linear patterns or long-term dependencies. This has led to a growing perception that deep learning models represent a superior solution for time-series forecasting tasks.

However, the literature also highlights important limitations. LSTM models are computationally expensive, sensitive to hyperparameter selection, and prone to overfitting when applied to smooth or trend-dominated economic time series. In addition, their black-box nature reduces interpretability, which can be problematic in economic and policy-related applications. Several studies note that performance gains reported for deep learning models are often influenced by data preprocessing choices rather than model architecture alone.

2.4 Regression-Based Forecasting and Feature Engineering

Regression-based forecasting methods represent an alternative approach that has received comparatively less attention in recent house price forecasting literature. Linear Regression models, when combined with carefully engineered lag features, can explicitly incorporate temporal dependency without relying on complex model structures.

Lag-based regression approaches have been shown to perform competitively in forecasting tasks involving stable economic indicators, particularly when historical values exhibit strong autocorrelation. By transforming temporal dependence into explanatory variables, these models can capture trend continuation and short-term dynamics in a transparent and computationally efficient manner.

Compared to deep learning approaches, regression-based models offer advantages in interpretability, stability, and ease of implementation. However, their effectiveness depends heavily on the quality of feature engineering and the selection of appropriate lag structures. As a result, their potential is often underestimated when studies prioritize algorithmic complexity over data representation.

2.5 Research Gap

Although extensive research has been conducted on house price forecasting, several limitations remain evident. Many studies evaluate models using inconsistent preprocessing pipelines, differing validation strategies, or selective performance metrics, which undermines the reliability of comparative conclusions. In addition, simpler regression-based approaches are frequently excluded from analysis, despite evidence suggesting that they can perform well on long-term economic time series.

There is therefore a lack of controlled and transparent comparisons that assess classical statistical models, deep learning approaches, and regression-based methods within a unified experimental framework. This gap is particularly relevant for national-level UK house price data, which exhibits strong trends and predictable seasonal patterns. Addressing this limitation requires systematic evaluation under consistent conditions to determine whether increased model complexity genuinely delivers superior forecasting performance.

3. Dataset, Exploratory Analysis and Preprocessing

This section states the dataset used in the study, its source, and the rationale for its selection. It also outlines the initial exploration and preparation steps required to ensure the data is suitable for forecasting. Given the sensitivity of forecasting models to data quality and temporal structure, careful dataset selection and preprocessing are critical to achieving reliable and interpretable results.

3.1 Data Source and Description

The dataset used in this study is derived from the UK House Price Index (UK HPI), which provides monthly records of average residential property prices across the United Kingdom. The data spans the period from January 1995 to December 2024, offering a sufficiently long historical horizon for analyzing long-term trends, seasonal behavior, and temporal dependency. For the purpose of this study, national-level average house prices were selected, resulting in a univariate time-series indexed by time.

3.2 Exploratory Data Analysis

Exploratory analysis was conducted to understand the structural characteristics of the UK house price series prior to model development. Visual inspection of the time-series revealed a strong long-term upward trend, reflecting sustained growth in UK property values over the study period. In addition to the overall trend, recurring fluctuations were observed at regular annual intervals, indicating the presence of seasonality.

The series also exhibited a high degree of persistence, with current values strongly influenced by previous observations. This behavior suggests significant autocorrelation, supporting the use of lag-based features and sequential modelling approaches. Furthermore, price variability increased over time, with larger absolute changes occurring in more recent years, indicating non-constant variance.

These characteristics highlight key challenges for forecasting, including non-stationarity, trend dominance, and seasonal structure. They also informed subsequent modelling decisions, particularly the inclusion of seasonal components in statistical models and the use of historical windows for regression and deep learning approaches.

Average Price Histograms

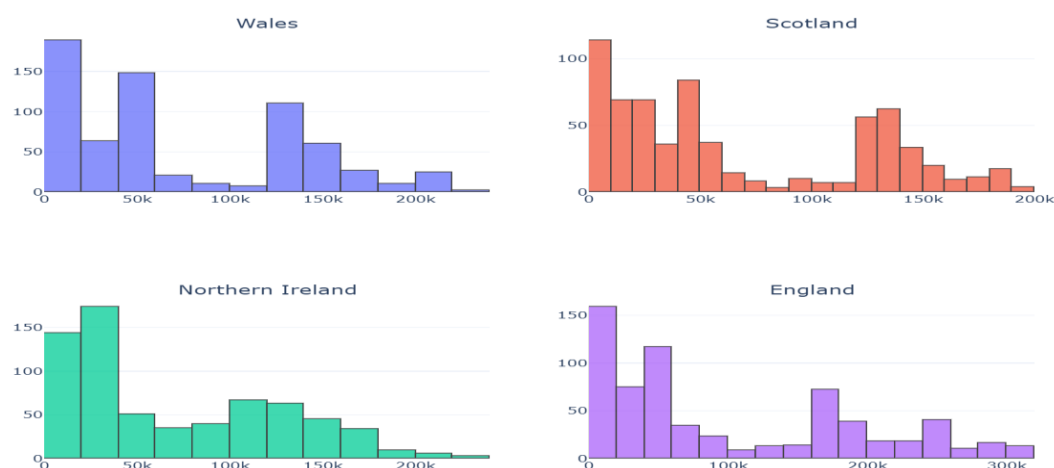


Figure 1: Average House price of different regions in Uk

3.3 Distributional Properties

Analysis of the distribution of house prices showed positive skewness, with lower values concentrated in earlier periods and a long right tail corresponding to higher prices in later years. This non-symmetric distribution reflects long-term market growth and suggests that variance stabilization techniques may be beneficial, particularly for models that assume homoscedasticity.

The presence of outlying high values in recent periods was not treated as noise, as these observations represent genuine market behaviour rather than data errors. Consequently, no outlier removal was performed, and all observations were retained to preserve the integrity of the time series.

3.4 Data Preprocessing

Based on insights from the exploratory analysis, several preprocessing steps were applied to ensure suitability for forecasting and consistency across models. First, the data was checked for missing values and structural gaps. As the UK HPI dataset is well maintained, no significant missing observations were identified within the selected time range.

To address non-stationarity, differencing was applied where required for statistical models, with stationarity formally assessed using the Augmented Dickey–Fuller test. This step ensured that the assumptions underlying ARIMA-based models were satisfied. For regression and deep learning models, the original series was retained, as these approaches do not require strict stationarity.

For models sensitive to scale, particularly the LSTM network, the data was normalised using min–max scaling to improve numerical stability and training efficiency. In addition, lagged features were engineered to explicitly represent temporal dependency for the Linear Regression model, allowing past price values to be used as predictors.

A chronological train–test split was applied to preserve temporal order, with earlier observations used for training and more recent data reserved for evaluation. This strategy prevents information leakage and reflects realistic forecasting conditions.

3.4.1 Trend and Seasonality Analysis

Visual inspection of the time-series revealed a clear long-term upward trend in average UK house prices over the study period, reflecting sustained market growth. In addition to the overall trend, recurring seasonal fluctuations were observed at regular yearly intervals, indicating the presence of seasonality in the data. These patterns justified the inclusion of seasonal modelling techniques such as SARIMA and supported the use of lag-based features for regression and deep learning models.

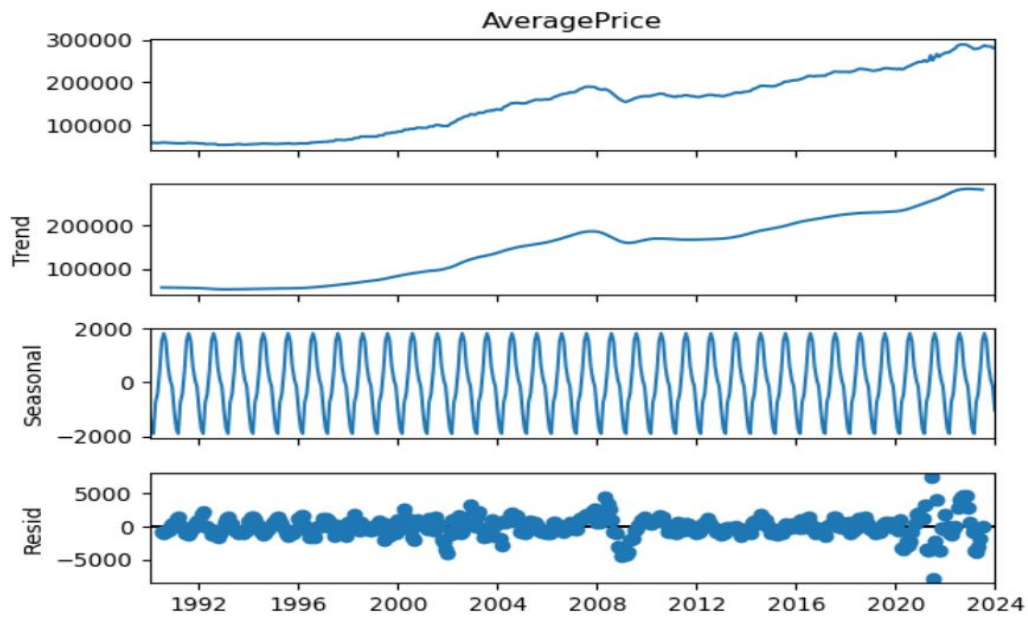


Figure 2: Trend and seasonality

3.4.2 Distribution of House Prices

Analysis of the price distribution showed positive skewness, with a concentration of lower values in earlier periods and a long tail corresponding to higher prices in recent years. This non-normal distribution further supported the application of transformation techniques to stabilize variance and improve model performance, particularly for statistical and regression-based approaches.

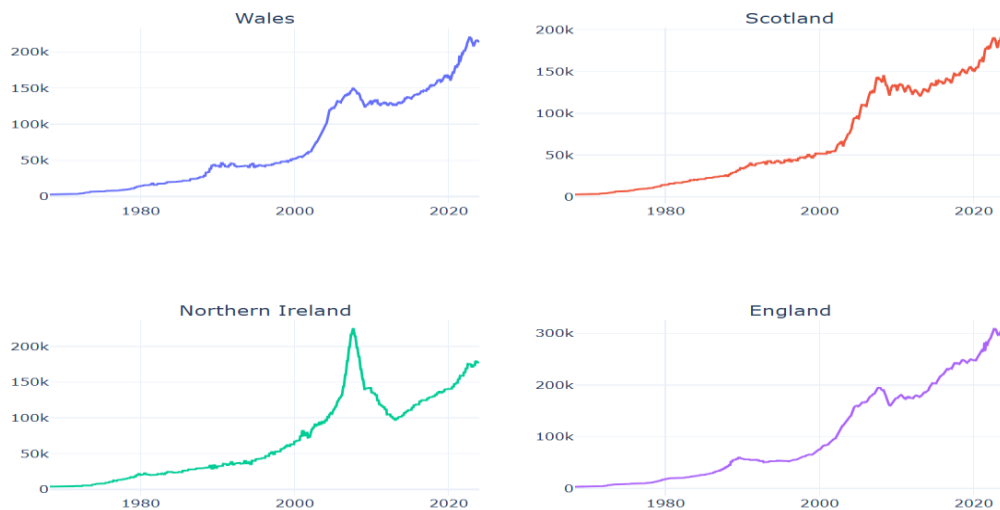


Figure 3: Distribution of House Prices

4. Ethical Considerations

This project was conducted in accordance with the ethical guidelines of the University of Hertfordshire and relevant data protection regulations. As the study relies exclusively on publicly available, aggregated data and does not involve human participants, personal identifiers, or sensitive information, the ethical risk associated with the project is minimal. Nevertheless, ethical considerations relating to data privacy, regulatory compliance, data usage, and academic integrity were carefully addressed throughout the research process.

5. Methodology

This section describes the methodological framework adopted to forecast UK house prices and to compare the performance of different forecasting models. A structured and time-aware approach was followed to ensure that all models were trained, tested, and evaluated under consistent conditions. The methodology includes dataset preparation, temporal data splitting, model development, and evaluation using standard forecasting metrics. Emphasis was placed on transparency, reproducibility, and fair comparison across model families.

5.1 Overview of the Modelling Framework

The modelling framework adopted in this study follows a sequential pipeline consisting of data preprocessing, exploratory analysis, model implementation, and performance evaluation. The workflow begins with preparing the monthly UK house price time series, followed by transforming the data to satisfy model assumptions. Multiple forecasting models—ARIMA, SARIMA, LSTM, and Linear Regression, were then implemented independently using the same underlying dataset.

All models were trained using historical data only and evaluated on unseen future observations to simulate real-world forecasting conditions. Performance was assessed using consistent error metrics to allow direct comparison across models. The entire modelling process was implemented using Python, with version control maintained through GitHub to ensure reproducibility.

5.2 Time-Aware Train–Test Split Strategy

Given the sequential nature of time-series data, a time-aware train–test split strategy was employed instead of random sampling. The dataset was divided chronologically, with earlier observations used for model training and the most recent observations reserved for testing. This approach prevents data leakage and ensures that future information does not influence model training.

5.3 Model Development Model Development

This subsection describes the implementation of each forecasting model used in the study. Each model was developed independently but evaluated using identical datasets and metrics to ensure methodological consistency.

5.4 ARIMA Model Implementation

The Autoregressive Integrated Moving Average (ARIMA) model was implemented as a baseline classical time-series forecasting approach. Prior to model fitting, the house price series was transformed and differenced where necessary to achieve stationarity. Model parameters were selected based on autocorrelation and partial autocorrelation analysis, along with iterative testing to minimize forecast error.

The ARIMA model was trained using historical data and used to generate forecasts over the test period. Its performance provided a benchmark for evaluating more advanced models.

5.5 SARIMA Model Implementation

Seasonal behaviors identified during exploratory analysis was addressed through the use of a Seasonal ARIMA (SARIMA) model. This approach augments the standard ARIMA framework by explicitly modelling repeating temporal patterns, enabling the representation of regular annual fluctuations in house prices. Seasonal orders were determined in line with the periodic structure observed in the time series rather than through arbitrary specification. To ensure comparability, the SARIMA model was trained and tested using the same chronological data split applied to the non-seasonal ARIMA model.

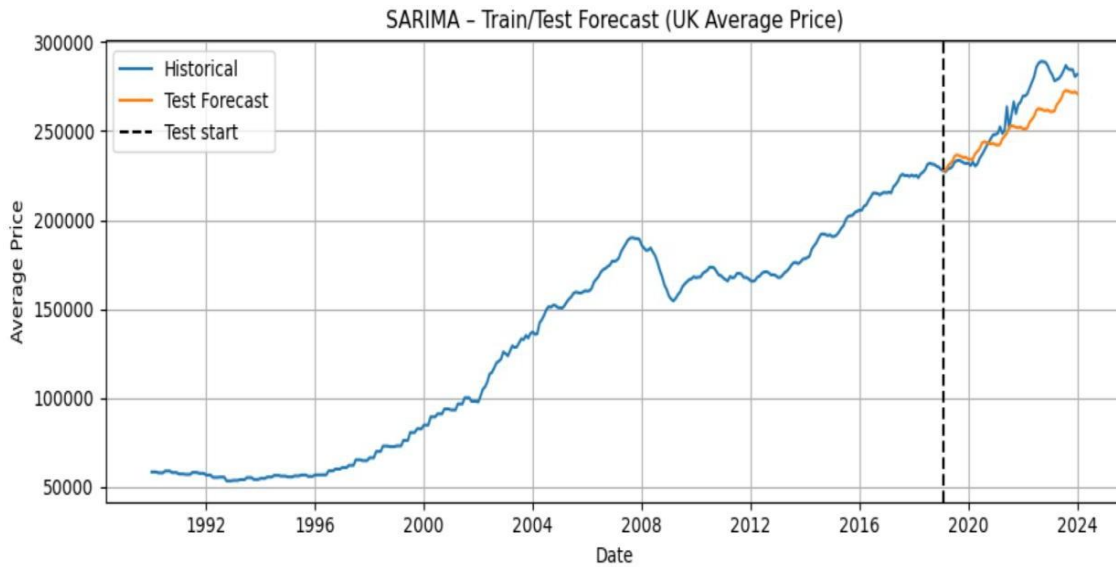


Figure 4: SARIMA Train- Test Results

5.6 LSTM Model Architecture and Training

A LSTM model was employed to capture temporal patterns in the house price series that cannot be represented effectively using linear assumptions. Prior to model training, the data was normalized and restructured into fixed-length sequences so that past observations could be used as inputs for future value prediction. The network design comprised an LSTM-based sequence learning component followed by a fully connected output layer responsible for generating forecasts. Model training was conducted using historical sequences, with

convergence assessed through both training and validation loss trajectories. To ensure stable generalization, model complexity and training behaviour were carefully controlled to reduce the risk of overfitting.

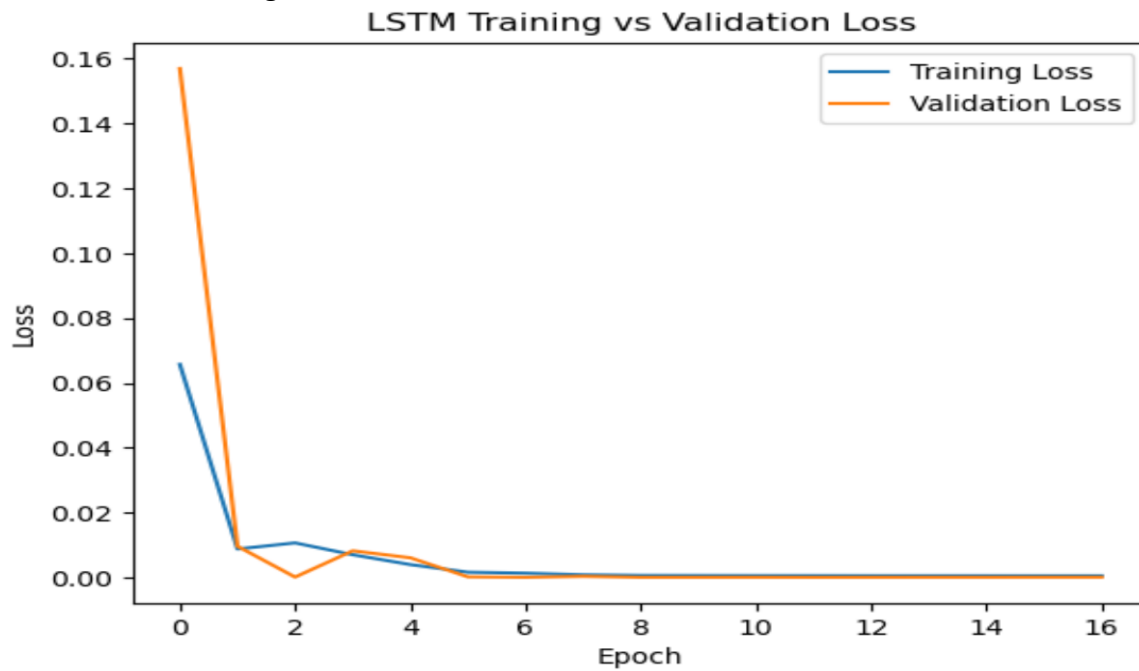


Figure 5: LSTM Training and validation Loss

5.7 Linear Regression with Lagged Features

A Linear Regression model was implemented using engineered lag features derived from historical house price values. These lag features allow the model to capture temporal dependencies by explicitly incorporating past observations as input variables.

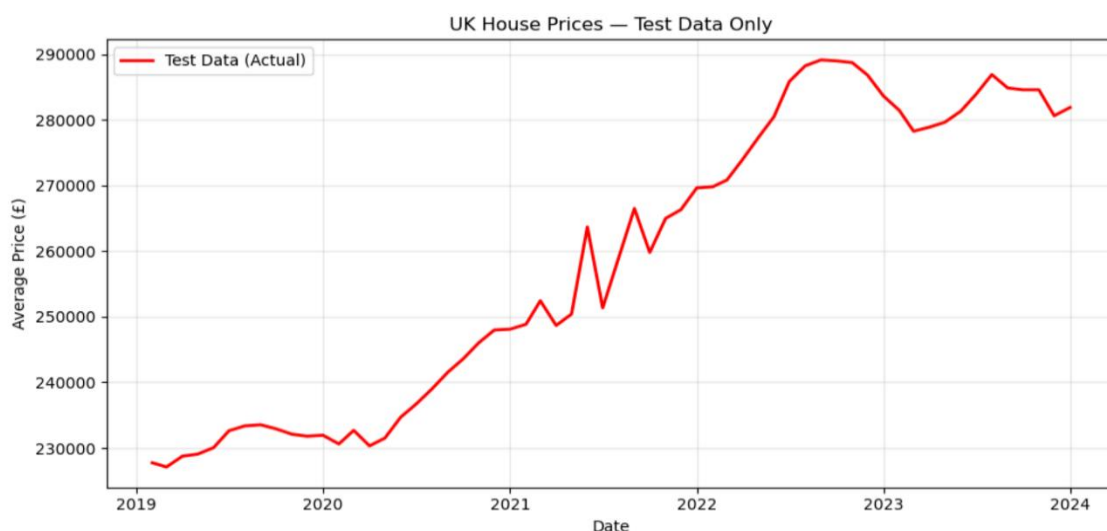


Figure 6: Linear Regression Test data

The number of lagged features was selected based on autocorrelation analysis and empirical

testing. Despite its simplicity, this approach provided a strong and interpretable benchmark. The model was trained using the same time-aware split and evaluated using identical metrics to ensure comparability with other models.

6. Results

This section presents the forecasting results obtained from each model implemented in the study. All models were evaluated using the same time-aware train–test split and assessed using standard forecasting metrics to ensure fair comparison. The results are reported individually for each model before comparative analysis in the subsequent discussion section.

6.1 ARIMA Model Results

The ARIMA model was used as a baseline classical time-series forecasting approach. After applying differencing to achieve stationarity, the model was trained on the historical UK house price data and used to generate forecasts for the test period. The ARIMA forecasts were able to capture short-term price movements but showed limited capability in modelling longer-term trends accurately. Forecast errors increased as the prediction horizon extended, indicating reduced robustness over time. Evaluation results showed relatively high RMSE and MAE values compared to other models, reflecting weaker predictive accuracy. These results highlight the limitations of ARIMA when applied to long-term, trend-dominated housing market data.

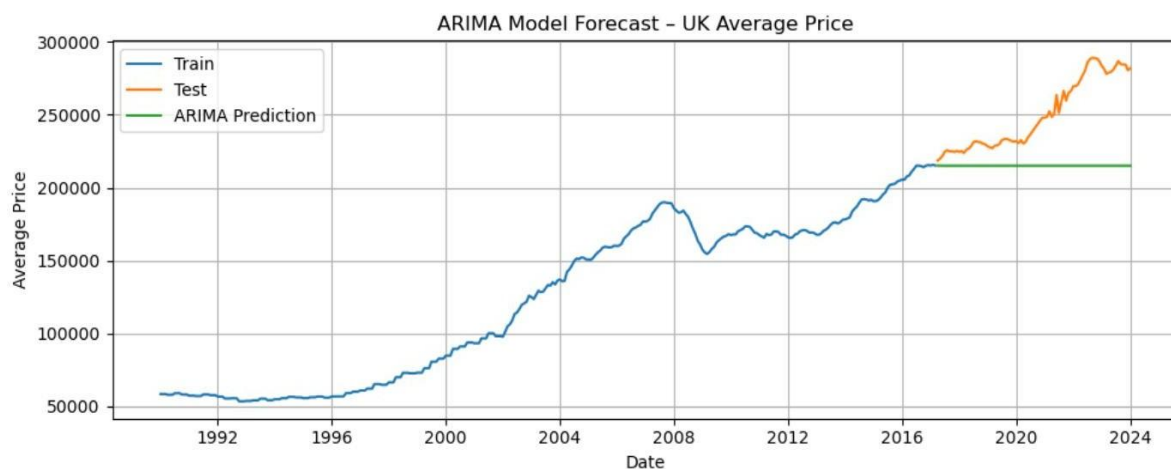


Figure 7: ARIMA Forecast Results

6.2 Augmented Dickey-Fuller test

The stationarity of the UK house price series was evaluated using the Augmented Dickey–Fuller test. The test produced a statistic of **0.9407** with an associated **p-value of 0.9936**, indicating that the null hypothesis of a unit root could not be rejected. The test was conducted using **16 lagged terms** and was based on **392 observations**.

The calculated test statistic exceeds the critical thresholds at all conventional significance levels, with critical values of **–3.4471 (1%)**, **–2.8689 (5%)**, and **–2.5707 (10%)**. These results confirm that the original house price series is non-stationary and requires transformation prior to the application of differenced time-series models.

6.3 SARIMA Model Results

By incorporating seasonal structure, the SARIMA model improved upon the baseline ARIMA approach by producing smoother forecasts that better reflected recurring annual patterns in the data. This resulted in reduced forecast errors during periods of pronounced cyclic behavior. However, despite its improved accuracy, SARIMA remained limited by its linear formulation, with forecast divergence increasing over longer horizons. Overall, SARIMA delivered more stable and realistic predictions than the non-seasonal ARIMA model.

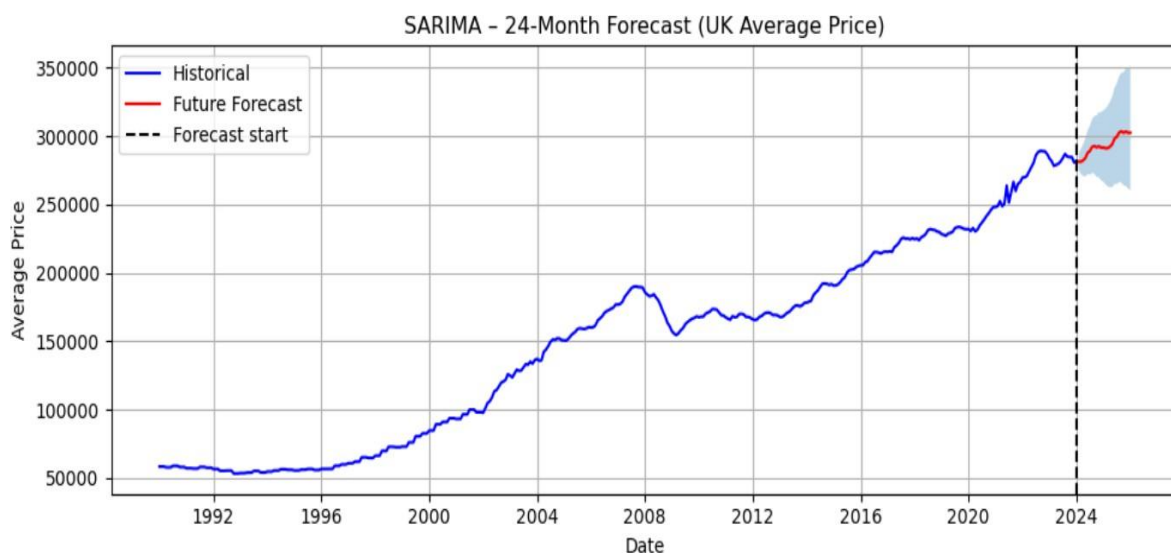


Figure 8: SARIMA Forecast Results

6.4 LSTM Model Results

The LSTM model learned temporal patterns from sequential house price data and converged quickly during training, with minimal divergence between training and validation loss. Forecasts closely matched observed test values, capturing both long-term trends and short-term fluctuations. Compared to classical time-series models, the LSTM achieved substantially lower RMSE and MAE, demonstrating its effectiveness in modelling complex temporal behavior.

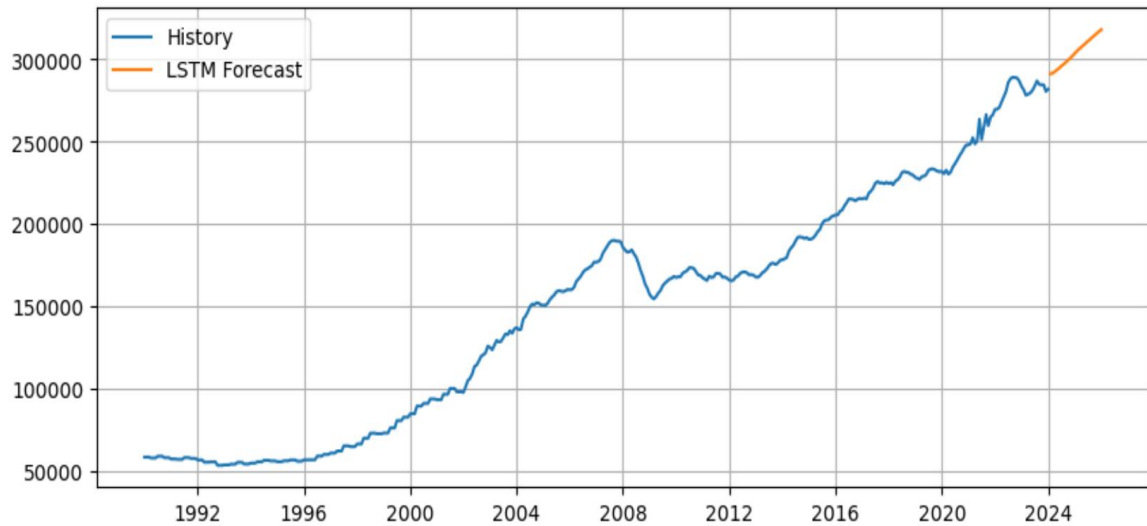


Figure 9: LSTM Forecast Results

6.5 Linear regression

The Linear Regression model with lagged features delivered the most accurate and consistent forecasts across all approaches. By using past price values directly as predictors, the model captured temporal structure effectively and produced smooth forecasts closely aligned with observed test data. It achieved the lowest RMSE, MAE, and MAPE, outperforming both statistical and deep learning models. This result highlights the impact of feature design over model complexity in long-term house price forecasting.

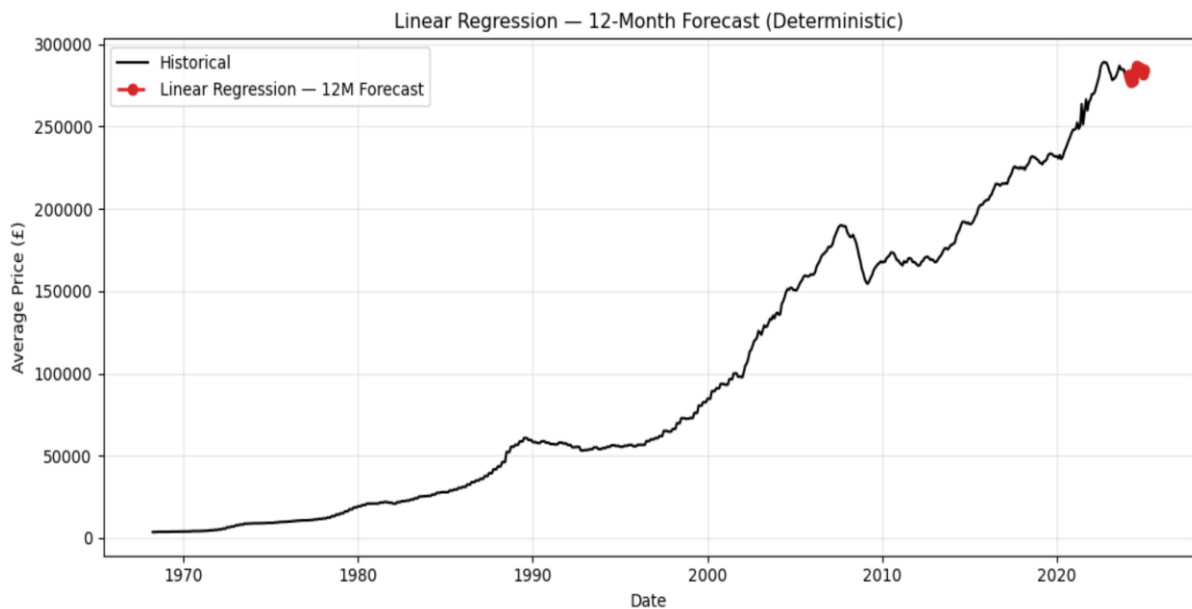


Figure 10: Linear Regression Forecast Results

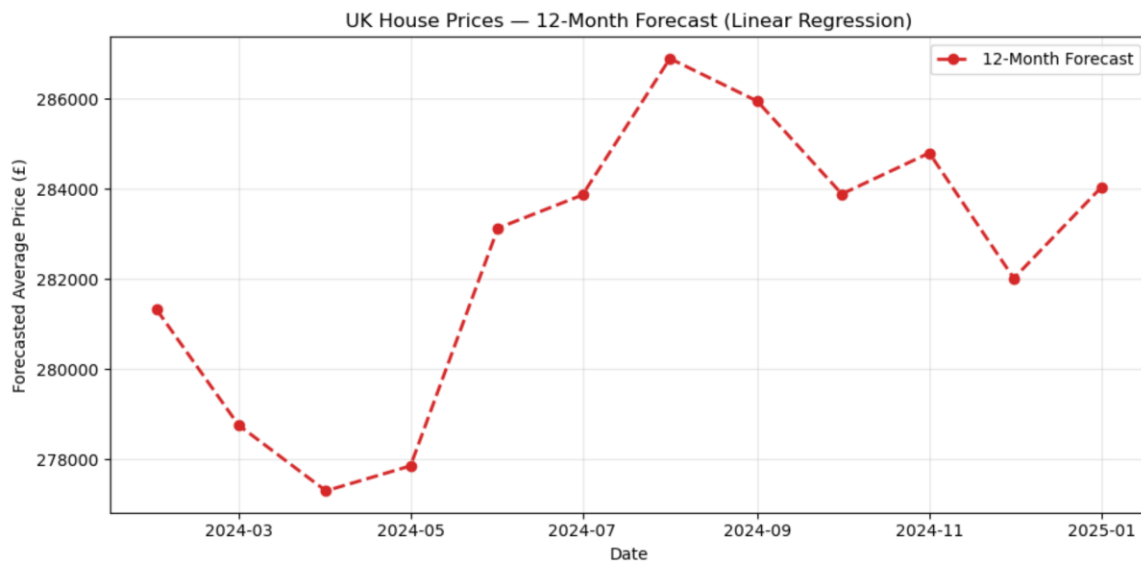


Figure 11: Linear Regression forecast results (12 months of data)

6.6 Hyperparameter Tuning of the LSTM Model

Hyperparameter tuning was performed for the LSTM model to improve forecasting performance while maintaining model stability and reproducibility. Given the highly autocorrelated and trend-dominated nature of UK house price data, the tuning process was intentionally constrained to avoid excessive model complexity and overfitting.

Hyperparameters Considered

The following LSTM hyperparameters were selected for tuning due to their direct influence on sequence learning and optimization behavior:

- **Sequence length (window size):** determines the number of past time steps used as input.
- **Number of LSTM units:** controls the model's representational capacity.
- **Learning rate (Adam optimizer):** affects the speed and stability of convergence during training.

Optimal Hyperparameter Configuration

The hyperparameter tuning process resulted in the following optimal configuration:

- **Sequence length (window size):** 12 months
- **Number of LSTM units:** 64
- **Learning rate:** 0.001 (Adam optimiser)

Results:

Window Size : 12
 LSTM Units : 64
 Learning Rate : 0.001
 Best RMSE : 5669.23

LSTM (Tuned) RMSE: 11051.30

LSTM (Tuned) R^2 : 0.7788

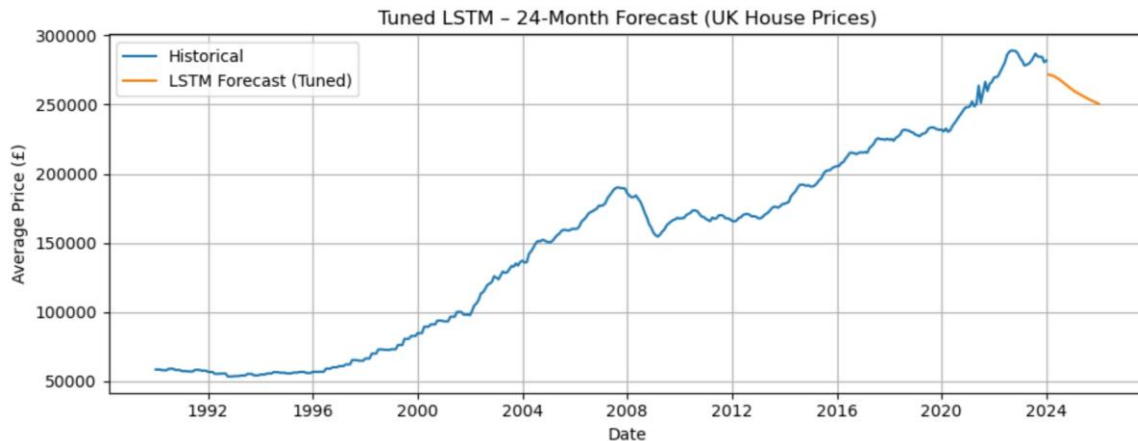


Figure 12: Hyperparameter Tuning LSTM Results

7. Evaluation Metrics

To objectively assess and compare the forecasting performance of all models, multiple evaluation metrics were employed. Using more than one metric ensures that model accuracy is examined from different perspectives, including absolute error magnitude, relative error, and explained variance. All metrics were calculated on the same test dataset to maintain consistency and fairness across model comparisons.

7.1 Root Mean Squared Error (RMSE)

RMSE quantifies the typical magnitude of forecasting errors by giving greater weight to larger deviations between predicted and observed values, making it suitable for assessing models where large errors are particularly costly.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

Where:

y_i = actual values

$\hat{y}_i$ = predicted values

n = number of observations

In this study, RMSE was used as the primary accuracy metric. The Linear Regression model achieved the lowest RMSE, indicating the highest overall forecasting accuracy. The LSTM model produced substantially lower RMSE than the ARIMA and SARIMA models, while ARIMA recorded the highest RMSE, reflecting weaker predictive performance.

ARIMA: 42043.53

SARIMA: 13274.05

LSTM: 2734594.40

Linear Regression: 3294.97

7.2 Mean Absolute Error (MAE)

MAE represents the average size of prediction errors by measuring absolute differences between forecasts and observed values, without placing additional emphasis on extreme deviations.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

MAE measures the average magnitude of forecasting errors and is easier to interpret than RMSE.

Results:

ARIMA: 34721.14

SARIMA: 10538.45

LSTM: 2734591.61

Linear Regression: 2232.14

7.3 Mean Squared Error (MSE)

MSE measures the average squared difference between predicted and actual values, assigning greater penalty to larger errors and providing an indication of overall prediction deviation.

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Results:

ARIMA: 1767658468.08

SARIMA: 176200544.94385925

LSTM: 7478006574813.41

Linear Regression: 10856884.08

7.4 R-squared (R^2)

R-squared (R^2), indicates how much of the variation in the observed values is explained by the model's predictions, providing a measure of overall explanatory strength.

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

ARIMA: -2.145
SARIMA: 0.6379
LSTM: -18259.60
Linear Regression: 0.9776

8. Analysis and Discussion

Clear differences were observed across the four forecasting approaches. ARIMA produced the weakest results, with high error values and a negative R^2 , indicating poor representation of long-term house price growth despite differencing. SARIMA improved accuracy by accounting for seasonal effects but remained limited by its linear structure. The LSTM model achieved substantially better performance, generating forecasts that closely followed observed price movements and reduced error metrics. However, the strongest overall results were obtained from the Linear Regression model with lagged features, which delivered the highest accuracy and stability across all metrics. This outcome indicates that effective feature representation was more influential than model complexity for this dataset.

8.1 Comparative Model Performance Analysis

The comparative analysis reveals clear performance differences across the four forecasting approaches. Classical time-series models, particularly ARIMA, exhibited the weakest performance, as evidenced by high error values and a negative R-squared score. This indicates that ARIMA struggled to capture the long-term growth structure of UK house prices, even after differencing to achieve stationarity. While SARIMA improved upon ARIMA by modelling seasonal effects, its linear structure limited its ability to adapt to complex temporal dynamics, resulting in moderate but still suboptimal accuracy.

The Linear Regression model achieved the best overall performance across all evaluation metrics. By explicitly incorporating historical price values through lag features, the model effectively captured temporal dependencies without introducing unnecessary complexity. Its superior accuracy and stability suggest that, for this dataset, structured feature representation played a more critical role than model sophistication.

8.2 Discussion of Results in Relation to Literature

The results broadly support existing studies that report strong performance from LSTM models compared to classical statistical approaches, particularly ARIMA and SARIMA. However, the superior performance of the Linear Regression model challenges the common assumption that greater model complexity guarantees higher forecasting accuracy. This outcome reinforces research that highlights the importance of feature representation and data structure over algorithmic sophistication. It also suggests that simpler models are often undervalued in comparative studies, despite their effectiveness on stable economic time-series data when evaluated under consistent experimental conditions.

8.3 Interpretation of Forecasting Behaviour

Differences in forecasting performance can be explained by the structural properties of the UK house price series, which is characterized by strong long-term growth, smooth transitions, and regular seasonal patterns. These features favor models that efficiently exploit historical dependence. ARIMA and SARIMA, despite their theoretical foundations, rely on linear assumptions and differencing that limit their ability to extend long-term trends. The LSTM model captured both trend and short-term variation but required careful tuning to perform consistently. In contrast, the Linear Regression model directly encoded temporal dependence through lagged features, producing stable forecasts that extended naturally over time. These findings suggest that effective representation of temporal information is more influential than model complexity alone in long-term forecasting tasks.

9. Conclusion and Future Work

This chapter summarizes the key outcomes of the project, acknowledges its limitations, and identifies opportunities for future research. The conclusions are drawn directly from the empirical results and analysis presented in previous sections.

9.1 Summary of Key Findings

This study compared statistical, deep learning, and regression-based models for forecasting UK house prices using monthly data from 1995 to 2024 under a unified evaluation framework. ARIMA showed limited accuracy on long-term housing data, while SARIMA achieved moderate improvement by modelling seasonality but remained restricted by linear assumptions. The LSTM model delivered substantially better performance by capturing complex temporal patterns. However, the Linear Regression model with lagged features produced the most accurate and stable forecasts across all metrics. These results indicate that well-designed feature representation and consistent methodology can be more influential than model complexity in time-series forecasting.

9.2 Future Research Directions

Future research could extend this work by developing region-specific forecasting models to capture geographical differences in UK housing markets. Incorporating macroeconomic and policy-related variables may further improve predictive accuracy and provide richer insights into market behavior. Hybrid modelling approaches that combine statistical methods with machine learning or deep learning techniques could also be explored to leverage the strengths of multiple model families. Additionally, investigating alternative neural network architectures and advanced feature engineering strategies may enhance long-term forecasting performance. These extensions would contribute to a more comprehensive and practical understanding of house price forecasting.

10. References

Hoang, T., 2024. Time series analysis on Connecticut housing market prices. SSRN Electronic Journal, SSRN Paper No. 5057723. [online] Available at: <<https://ssrn.com/abstract=5057723>> .

Azlan, A., Yusof, Y. and Mohsin, M.F.M., 2019. Determining the impact of window length on time series forecasting using deep learning. International Journal of Advanced Computer Research, 9(44), pp.260–267.

Pattanayak, R.M., Behera, H.S. and Panigrahi, S., 2021. A novel probabilistic intuitionistic fuzzy set based model for high order fuzzy time series forecasting. Engineering Applications of Artificial Intelligence, 99, p.104136. <https://doi.org/10.1016/j.engappai.2020.104136>

Box, G.E.P., Jenkins, G.M., Reinsel, G.C. and Ljung, G.M., 2015. Time series analysis: Forecasting and control. 5th ed. Hoboken, NJ: John Wiley & Sons.

Fathi, O., 2019. Time series forecasting using a hybrid ARIMA and LSTM model. SSRN Electronic Journal. [online] Available at: [Google Scholar](#)

Chen, P., Niu, A., Liu, D., Jiang, W. and Ma, B., 2018. Time series forecasting of temperatures using SARIMA: An example from Nanjing. IOP Conference Series: Materials Science and Engineering, 452, p.052024. IOP Publishing. Available at: [Google Scholar Link](#)

Siami-Namini, S., Tavakoli, N. and Namin, A.S., 2018. A comparison of ARIMA and LSTM in forecasting time series. In: Proceedings of the International Conference on Machine Learning and Applications (ICMLA). IEEE, pp.1394–1401. Available at: [Google Scholar Link](#)

Chang, Z., Zhang, Y. and Chen, W., 2018. Effective Adam-optimized LSTM neural network for electricity price forecasting. In: 2018 IEEE 9th International Conference on Software Engineering and Service Science (ICSESS). IEEE, pp.245–248.

Dubey, A.K., Kumar, A., García-Díaz, V., Sharma, A.K. and Kanhaiya, K., 2021. Study and analysis of SARIMA and LSTM in forecasting time series data. Sustainable Energy Technologies and Assessments, 47, p.101415.

Christopher, L., Goh, G. and Lee, M., 2004. House price prediction: Hedonic price model vs. artificial neural network. American Journal of Applied Sciences, 1(1), pp.1–10.

Ramachandran, R., Rajeev, D.C., Krishnan, S.G. and Subathra, P., 2015. Deep learning – An overview. International Journal of Applied Engineering Research, 10(10), Available at : [Science Direct](#)

Park, B. and Bae, J.K., 2015. Using machine learning algorithms for housing price prediction: The case of Fairfax County, Virginia housing data. Expert Systems with Applications, 42(6).

Majidpour, M., Qiu, C., Chu, P., Gadh, R. and Pota, H.R., 2015. Fast prediction for sparse time series: Demand forecast of EV charging stations for cell phone applications. IEEE Transactions on Industrial Informatics, 11(1), pp.242–250.

Kotsiantis, S.B., Kanellopoulos, D. and Pintelas, P.E., 2006. Data preprocessing for supervised learning. International Journal of Computer Science, 1(2), pp.111–117.

Vaishnavi, A.V.S.S.P.L., Raghavendra, G.G.K., Jilan, M., Chowdary, A.P., Singh, R. and Karthikeyan, C., 2024. Exploring house price forecasting through machine learning and data preprocessing. In: 2024 4th International Conference on Pervasive Computing and Social Networking (ICPCSN). IEEE. <https://doi.org/10.1109/ICPCSN62568.2024.00055>

Truong, Q., Nguyen, M., Dang, H. and Mei, B., 2020. Housing price prediction via improved machine learning techniques. Procedia Computer Science, 174, pp.433–442.

Ho, W.K., Tang, B.S. and Wong, S.W., 2021. Predicting property prices with machine learning algorithms. Journal of Property Research, 38(1), pp.48–70.
<https://doi.org/10.1080/09599916.2020.1832558>

Han, J., Kamber, M. and Pei, J., 2012. Data mining: Concepts and techniques. 3rd ed. Waltham, MA: Morgan Kaufmann.

11. Appendix

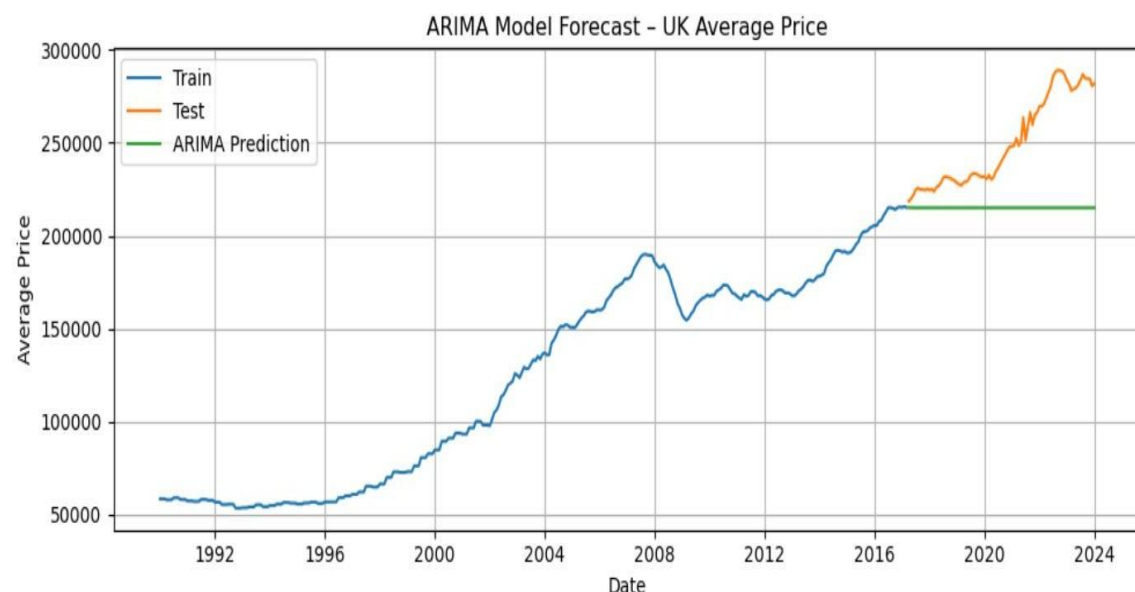


Figure 13: ARIMA Forecast Results

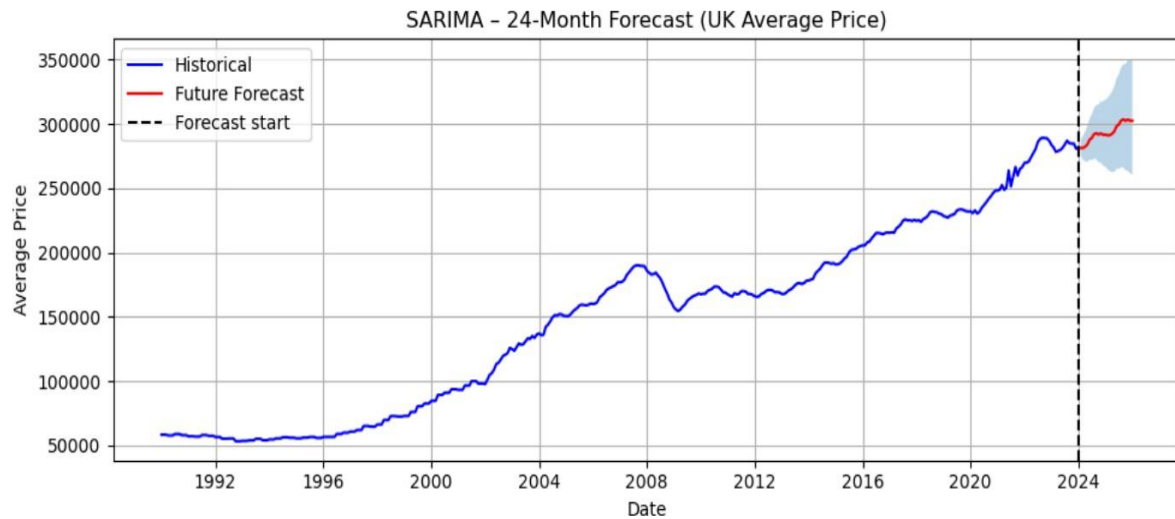


Figure 14: SARIMA Forecast Results

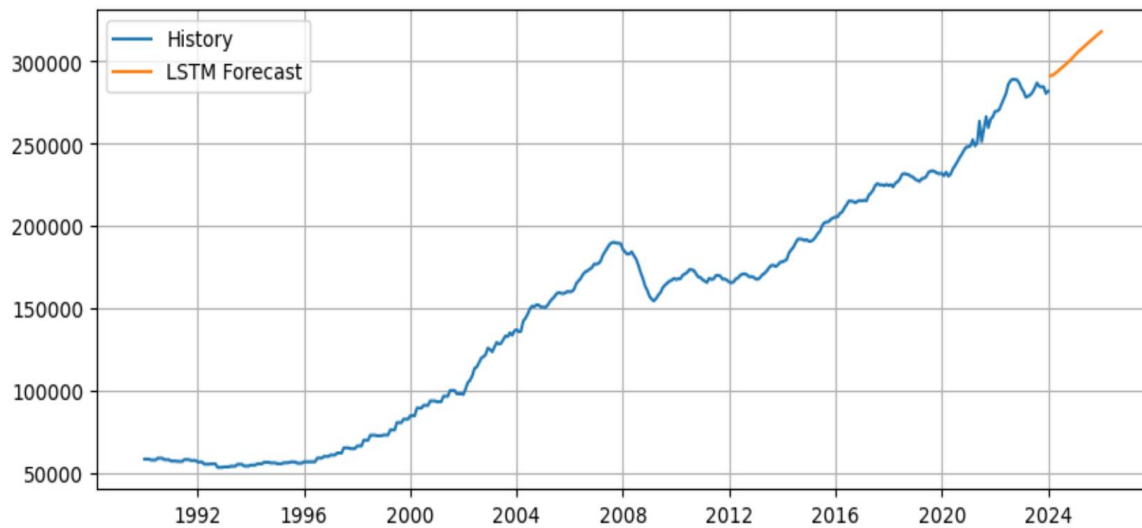


Figure 15: LSTM Forecast Results

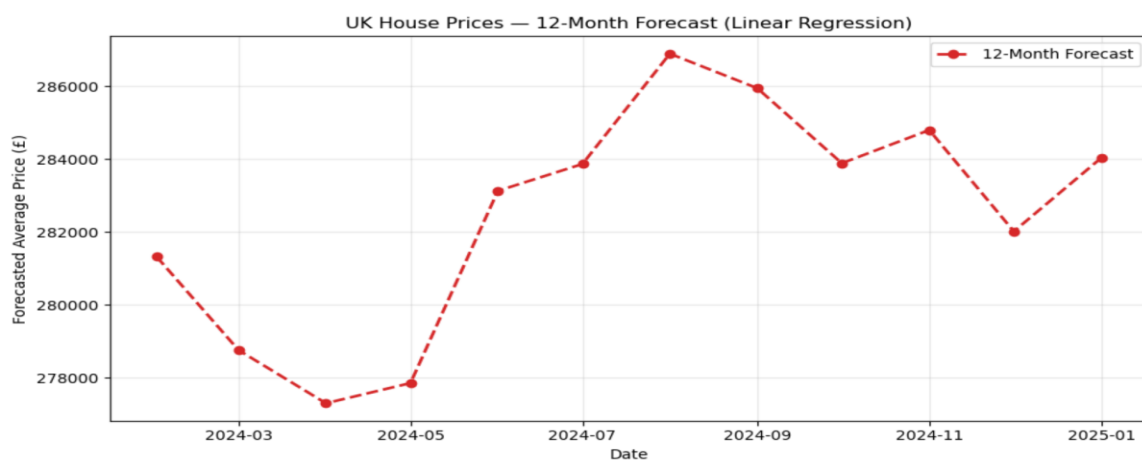


Figure 16: Linear Regression Forecast Results